

Week 7 Recitation

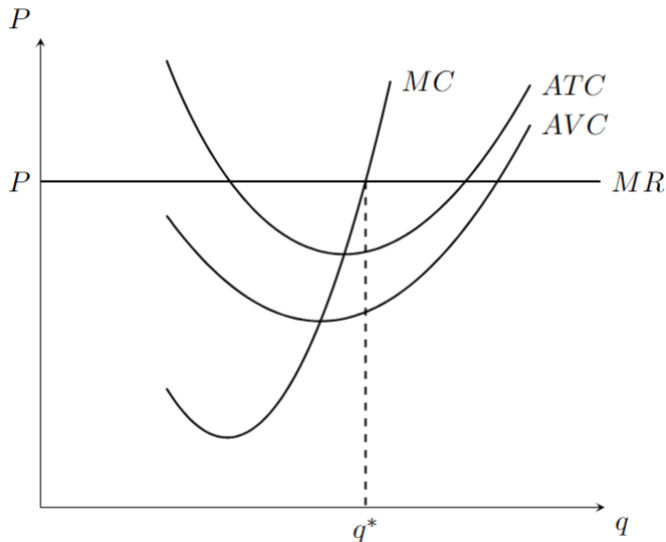
TA Suha Kim

ECON2001.01 (Prin. Micro.)

Feb 21, 2025

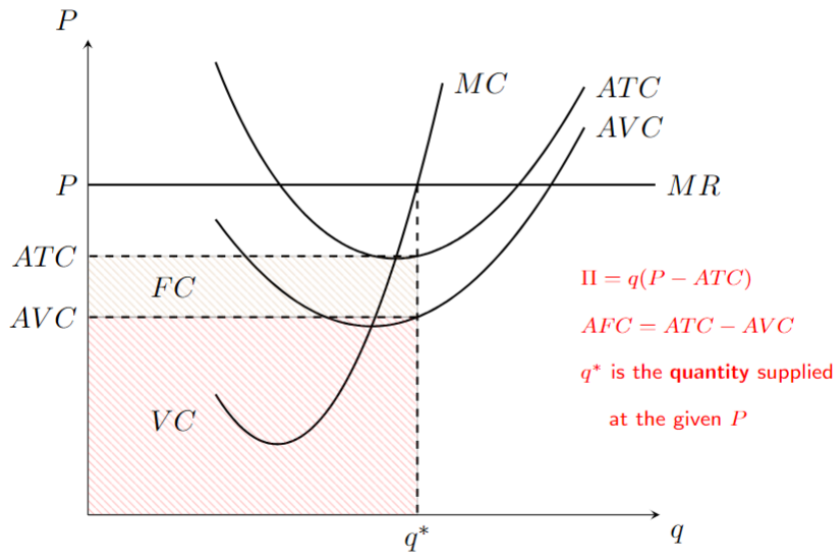
Seller's Problem

Seller's Decision - Optimal Q -supplied and Profit



Seller's Problem

Seller's Decision - Optimal Q-supplied and Profit



Seller's Problem

Seller's Decision - Shut-down

Shutdown is also a **short-run** decision.

- The firm is still operating as a business, but is choosing not to produce any output
- Because of fixed factors of production in the short-run, firms are not able to exit the market

If $\Pi = -FC$, the firm is indifferent between shutting down production and producing q

If $\Pi < -FC$, the firm maximizes its profit (minimizing the loss) by shutting down production

The shutdown point is where $P = MC = AVC$
: The minimum point of the AVC curve

Seller's Problem

Seller's Decision – Short run Supply curve

- Short run Supply curve = **MC curve**
- Price Elasticity of (Q-)supply
- Producer Surplus
 - Producer surplus (PS) is the difference between the **price** received from selling the good(s) and the minimum **willingness to accept** (WTA) for the good(s)
 - E.g., Sold a pair of shoes
A firm sold a pair of shoes for \$75
The firm would have been willing to accept \$50 for the pair
Therefore, the producer surplus for the one pair is,
 $PS = \$75 - \$50 = \$25$
 - the area below price & above supply (MC) upto the quantity sold

Seller's Problem

Long run Production & Cost

The **long run** is period of time in which **all factors of production are variable**

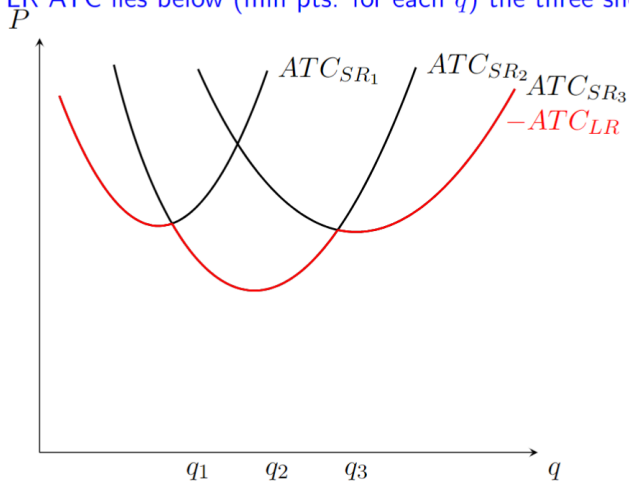
In the long-run, (individual seller)

- LR Total costs = LR Variable costs (So, $LR\ ATC = LR\ AVC$)
- The firm chooses to produce q , if and only if the profit is nonnegative (Otherwise, the firm will exit the market – Free entry/exit in the LR in a Perfectly competitive market!)
- A firm's LR supply is the MC curve from the minimum point of the LR ATC curve and up

Seller's Problem

Long run Production & Cost

The LR ATC lies below (min pts. for each q) the three short-run ATCs



Seller's Problem

Market supply in the LR

In the long-run equilibrium, all firms (assuming identical costs structures) will earn **zero economic profit**

- Why? – Remember, **Free entry / Free exit in the LR!**

This means that no firm has an incentive to exit the market nor do firms have an incentive to enter the market

If the market is not in equilibrium, then the dynamics of the market are as follows:

- ① If the profit is positive,
 - Positive profits will incentivize outside firms to enter the market
 - Market supply increases, lowering the market price
 - Firms continue to enter the market until $\text{Profit} = 0$
- ② If the profit is negative,
 - Existing firms in the market will be incentivized to exit the market
 - Market supply will decrease, increasing the market price
 - Firms continue to exit the market until $\text{Profit} = 0$

Seller's Problem

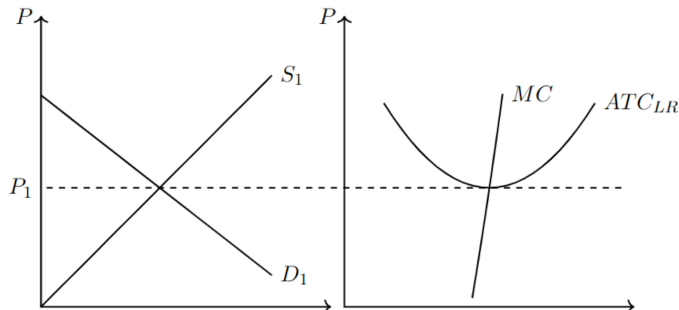
Market supply in the LR

In the long-run equilibrium, all firms (assuming identical cost structures) will earn **zero economic profit**

- Why? – Remember, **Free entry / Free exit in the LR!**

This means that no firm has an incentive to exit the market nor do firms have an incentive to enter the market

If the market is not in equilibrium, then the dynamics of the market are as follows:



Seller's Problem

Comparison SR vs. LR – Summary

- To maximize the profit, the quantity supplied is determined where $MR = MC$, and
 - If in the short-run, $\Pi \geq -FC$ ($P \geq \mathbf{AVC}$)
 - If in the long-run, $\Pi \geq 0$ ($P \geq \mathbf{ATC}$)
- The firm's supply curve is
 - In the short-run: the MC curve from the minimum point of the **AVC** curve on up
 - In the long-run: the MC curve from the minimum point of the **ATC** curve on up
- In the long-run **market equilibrium**, all firms in the market will earn **zero economic profit**
 - This is because of free (no barriers) entry and exit in and out of the market

Market Efficiency

Overview

Absent of a **market distortion**, equilibrium in a perfectly competitive market is **pareto efficient**

- Social surplus is maximized

The equilibrium price has **three distinct functions** in a perfectly competitive market: (Invisible Hands)

- ① Efficiently allocates resources across buyers and sellers
 - To max Social surplus, matches buyers with highest reservation value to seller with lowest reservation value
- ② Efficiently allocates resources across firms
 - $MC = MR \Rightarrow$ All firms produce their last unit at the same cost (i.e., **MCs are all equalized** by the efficient allocation)
- ③ Efficiently allocates resources across markets
 - LR equilibrium, all firms earn zero profit (No other outside options for firms that yield a greater return)
 - Price guides resources for the firms to produce at the lowest cost

Market Efficiency

Across Buyers and Sellers

About the point (1):

Matching buyers and sellers regarding their reservation values
⇒ Social Surplus is maximized?

Exhibit 7.1 Reservation Values of Buyers and Sellers in the iPhone Market

Quantities are only equal at \$40

Buyers	Res. Value (\$)	Cum. Q	Sellers	Res. Value (\$)	Cum. Q
Madeline	\$70	1	Tom	\$10	1
Katie	\$60	2	Mary	\$20	2
Sean	\$50	3	Jeff	\$30	3
Dave	\$40	4	Phil	\$40	4
Ian	\$30	5	Adam	\$50	5
Kim	\$20	6	Matt	\$60	6
Ty	\$10	7	Fiona	\$70	7

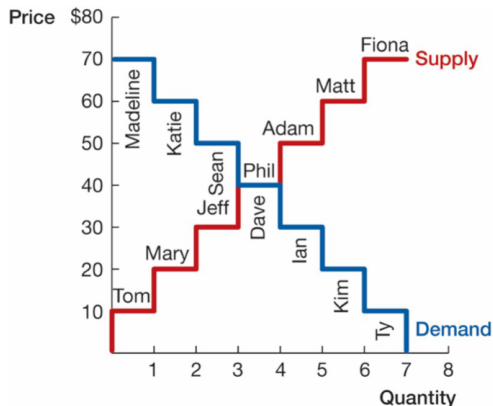
Market Efficiency

Across Buyers and Sellers

About the point (1):

Matching buyers and sellers regarding their reservation values

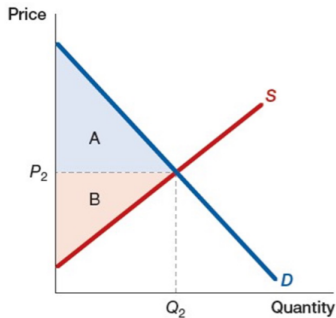
⇒ Social Surplus is maximized?



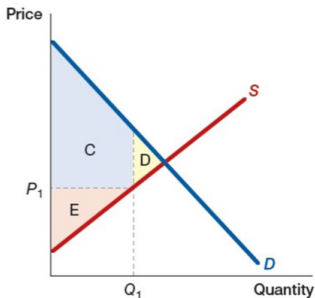
Market Efficiency

Dead Weight Loss

What happens when market intervention?



(a) Surplus in a Free Market



(b) Surplus with a Price Control